

Multiple Choice (10 marks)

1. Cells take in food for energy. The part of the cell that aids in digestion of the food is the lysosome. What is the main role of lysosomes in the process of food digestion?
 - (a) building proteins
 - (b) breaking down wastes
 - (c) controlling the activities of the cell
 - (d) converting energy from one form into another
2. Which of these activities will result in a different kind of material being formed?
 - (a) A nail is left outside and it rusts.
 - (b) A glass is dropped and it shatters into small pieces.
 - (c) A rubber band is stretched until it breaks.
 - (d) A pencil is sharpened to a point.
3. The same side of the Moon is always visible from Earth because
 - (a) the Moon does not rotate on its axis.
 - (b) the far side of the Moon faces Earth during the day.
 - (c) the Moon makes one rotation per revolution around Earth.
 - (d) the rate of rotation is the same for Earth and the Moon.
4. A statue and a table are both made of the same type of marble. Which of the following properties will most likely be the same for both of these objects?
 - (a) size
 - (b) shape
 - (c) weight
 - (d) hardness
5. Which statement best describes a structural adaptation that helps an organism survive in a desert?
 - (a) A bird builds its nest in a cactus.
 - (b) A rattlesnake shakes its tail when threatened.
 - (c) A lizard crawls under a rock for shelter.
 - (d) The ears of a jackrabbit release heat.

True / False (10 marks)

Answer True or False

6. True or False: Heat energy from the Sun is primarily transferred to Earth through conduction in the atmosphere.
7. True or False: A standardized taxonomic classification system is primarily used to determine the geographical habitats of all organisms.
8. True or False: The average distance between Earth and the Sun can be expressed as 149.6×10^5 kilometers.
9. True or False: When a battery-operated flashlight is turned on, electrical energy is transformed into light energy.
10. True or False: Designing an experiment that tests multiple types of popcorn with repeated trials is the best method to determine popped kernel yield.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the processes that contribute to the formation of a beach surface, including the role of natural elements such as water, wind, and sediment.
12. Discuss how the study of earthquakes contributes to our understanding of Earth's geological history and the processes that shape our planet over time.

End of Paper